Ahmedabad University

CSE400

CLIMATE AND WEATHER SYSTEMS

Problem Statement: *Probabilistic Precipitation Nowcasting*

M3 Submission

Section 2 - Group 2

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Probability Precipitation Nowcasting

- Extremely brief forecasting of precipitation (rainfall) is known as precipitation nowcasting, and is typically provided on a few-hour or hourly basis, based on radar data.
- Radar systems record rainfall in spatial grids whereby at every pixel, the amount of rainfall intensity at a particular point is recorded.
- The process of rainfall system is dynamic; it changes because of the dynamics in the atmosphere, wind flows, and clouds.
- Such variations render forecasting of rainfall uncertain and complicated in nature.
- Statistical modeling and physical modeling methods are applied to gain insight and simulation of the evolution of rainfall.
- Nowcasting radar systems monitor alterations in the movement and intensity of rainfall through movement estimation and time modeling.
- This project examines the evolution of rainfall with the help of a deterministic S-PROG based nowcasting framework.

Project Objective and Scope

- The aim of this project is to engage in the modeling and simulation of the short-term evolution of rainfall with the help of the techniques of radar-based nowcasting.
- The patterns of rainfall vary across space and time and therefore can not be predicted correctly in the short term.
- Rather, the system relies on the most recent radar rainfall data to make predictions of the intensity of the rainfall at time steps ahead.
- Fields of rainfall are processed and transformed in order to use rainfall as the subject of statistical modeling.
- The model can be used to simulate the changes of rainfall by applying motion estimation, cascade decomposition, and AR(2) temporal prediction.
- This deterministic integration gives a minimum model of probabilistic forecast of future rainfall.

Project System

- The project system carries out short-term rainfall nowcasting with the use of radar rainfall.
- Radar data is displayed in 2D grids of rains with each cell denoting the intensity of rain.
- Three successive rain-field fields R(t-2), R (t-1), R (t) are used as input to the system.
- The values of rainfalls are initially log-transformed to ensure that they stabilize the variability.
- S-PROG deterministic pipeline is based on motion estimation, cascade decomposition, and AR temporal modeling.
- The system produces future rain forecast fields over several steps in time.

Sources of Uncertainty

- Rainfall forecasting is not so definite since weather systems are unstable and unpredictable. The weather tends to change with time and thus, precision in growth rather challenging.
- Radar measurements can have measurement noise or errors in estimating the rainfall and this obstructs the precision of rainfall data.
- There are three processes through which rainfall systems can grow, decay and move spatially and as a result, the rainfall pattern can vary between time steps.
- Minor fluctuations in weather patterns may cause extensive difference in nature and place of rainfall.
- The uncertainty about these reasons provokes the adoption of the techniques of statistical and probabilistic modeling in rainfall prediction.

Key Random Variables and their Roles

- Rainfall Intensity R (x, y, t)

The amount of rain that falls at one point of space at time t, and it is the primary variable to nowcast.

- Log-Transformed Rainfall Y (x, y, t) = 10log10(R (x, y, t))

Applied to stabilize the variability in rainfall intensity and allow statistical modeling of the rainfall development.

- Rainfall Motion Field V

It uses rainfall images to calculate the rain pattern

- Variable of Exceedance Indicator.

It is used to determine whether expected rainfall is above threshold value, which is used in determining possible extreme rain patterns.

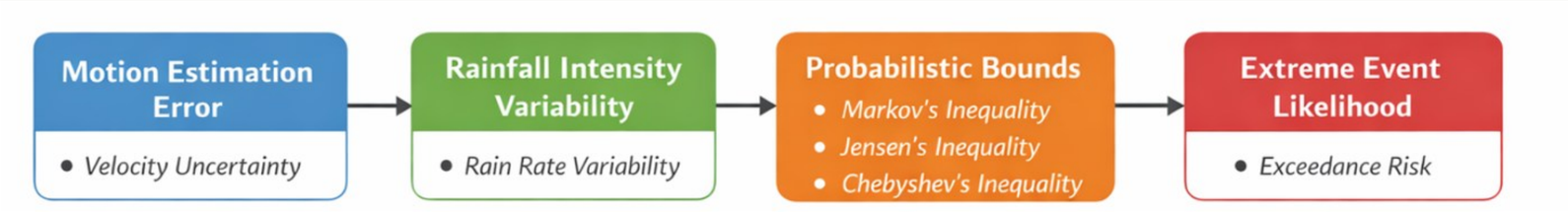
Probability Models and Assumptions

- The intensity of rainfall is considered to be log- normally behaving and therefore the values of rainfall are transformed to log scale.
- The values of rainfall are log-transformed making them more or less Gaussian, and thus subject to statistical modelling.
- A deterministic AR(2) time-varying model is used to model the evolution of the rainfall over time.
- Motion field is estimated based on radar images that represent the movement of rainfalls across space.
- Variability in rainfalls is present at various spatial scales and this is modeled by cascade decomposition.
- The assumptions enable the system to simulate the dynamics of rainfalls and retain a deterministic nowcasting structure.

Probabilistic Reasoning and Inference Logic

- INPUT: Radar observations as 2D rainfall grids (Matrix of pixels: each pixel = rainfall in mm/h)

  Three consecutive time steps: R(t-2), R(t-1), R(t). Typical resolution: 1 km, 5-minute intervals
- Rainfall forecasting contains uncertainty due to changing weather patterns.
- Rainfall fields are first log-transformed to stabilize variability.
- Motion estimation tracks the movement of rainfall patterns.
- Cascade decomposition separates rainfall into multiple spatial scales.
- Each scale is modeled using a deterministic AR(2) temporal model.
- Predictions are advected using the estimated motion field.
- Forecast rainfall is obtained by reconstructing all scales together.



Operationalizing Probabilistic Reasoning

- Implementation of a deterministic S-PROG forecasting pipeline
- Steps used in simulation:
  - Input rainfall fields R(t−2),R(t−1),R(t)
  - Log transformation
  - Motion estimation (Optical Flow)
  - Cascade decomposition (8 spatial scales)
  - Lagrangian transformation of rainfall fields
  - AR(2) parameter estimation
  - Deterministic AR(2) prediction
  - Advection of predicted rainfall fields
  - Reconstruction and inverse transform
- Forecast outputs generated for multiple time steps
- System performance evaluated using:
  - Forecast rainfall maps
  - Error comparison with observed rainfall
- Deterministic baseline established for future probabilistic comparison

Current Limitations and Conceptual Gaps

- The current model is fully deterministic
- Does not capture uncertainty in motion estimation
- AR(2) assumes linear temporal dependence
- Cascade decomposition may lose some spatial information
- Real rainfall systems may contain stochastic variability
- Model performance depends on quality of input rainfall data

Planned Refinements and Role Coordination

- Extend deterministic model into probabilistic forecasting framework
- Incorporate uncertainty in motion estimation
- Add stochastic perturbations to rainfall fields
- Improve rainfall variability modeling
- Validate model using larger rainfall datasets

Role Coordination:

- Rim and Dhruvi - Video/PPT
- Shine and Sharanya - Scribe
- Vamsi - Concept Map

Summary of Group-Level Understanding

- Rainfall forecasting modeled using deterministic S-PROG framework
- Key modeling components:
  - Log rainfall transformation
  - Motion estimation using optical flow
  - Cascade decomposition into spatial scales
  - AR(2) temporal prediction
  - Reconstruction of rainfall fields
- Current model provides deterministic baseline forecasts
- Future work will incorporate uncertainty and probabilistic modeling

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THANK YOU!